



JABchem



Not to be shared without the copyright holder's permission

Past Papers Higher Chemistry

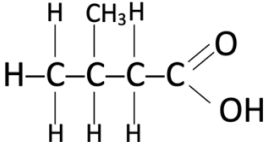
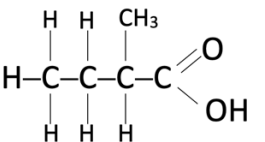
2024 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/150)	%	
A	101+	67%	29.7%
B	83+	55%	23.0%
C	65+	43%	21.6%
D	47+	31%	16.5%
No award	<47	<31%	9.2%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	15.8 /25	48.7 /95	19.4 /30

2024 Higher Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning																																																
1	C	53	☒ A Number of electrons in same outer electron shell does not decide the value of the first ionisation energy. ☒ B Neutrons in nucleus are not involved in the removal of electrons from outer shell ☒ C Chlorine has a greater nuclear charge which holds outer electrons more tightly as they are closer to nucleus. This increases the first ionisation energy of chlorine. ☒ D Mass of an atom has no influence on ionisation energies																																																
2	D	48	<table><tr><th colspan="6">State at 30°C</th><th colspan="6">Covalent Radii</th></tr><tr><th colspan="6">Melting Points for Group 1 Alkali Metals:</th><th colspan="6">Covalent Radii for Group 1 Alkali Metals:</th></tr><tr><td>Li</td><td>Na</td><td>K</td><td>Rb</td><td>Cs</td><td>Fr</td><td>Li</td><td>Na</td><td>K</td><td>Rb</td><td>Cs</td><td>Fr</td></tr><tr><td>181°C</td><td>98°C</td><td>63°C</td><td>39°C</td><td>28°C</td><td>less than 28°C</td><td>130pm</td><td>160pm</td><td>200pm</td><td>215pm</td><td>238pm</td><td>greater than 238pm</td></tr></table> As you go down group 1, the melting point of the Alkali Metals decreases. - melting point of Francium must be below the 28°C melting point of Caesium - If Francium has a melting point below 28°C then Francium will be a liquid at 30°C as it will have melted at 30°C. As you go down group 1, the covalent radii of the Alkali Metals increases. - An addition electron shell (energy level) is added to each member of the alkali metals going down Group 1 - Francium has one more electron shell than caesium so has a covalent radii greater than 238pm	State at 30°C						Covalent Radii						Melting Points for Group 1 Alkali Metals:						Covalent Radii for Group 1 Alkali Metals:						Li	Na	K	Rb	Cs	Fr	Li	Na	K	Rb	Cs	Fr	181°C	98°C	63°C	39°C	28°C	less than 28°C	130pm	160pm	200pm	215pm	238pm	greater than 238pm
State at 30°C						Covalent Radii																																													
Melting Points for Group 1 Alkali Metals:						Covalent Radii for Group 1 Alkali Metals:																																													
Li	Na	K	Rb	Cs	Fr	Li	Na	K	Rb	Cs	Fr																																								
181°C	98°C	63°C	39°C	28°C	less than 28°C	130pm	160pm	200pm	215pm	238pm	greater than 238pm																																								
3	C	62	☒ A C=O bonds are polar due to electronegativity difference of 0.8 CO₂ is a non-polar molecule overall due to linear shape ☒ B H-S bonds have an electronegativity difference of 0.4 and the bonds are not pure covalent as electrons are not shared completely evenly ☒ C P-H bonds are non-polar due to P and H atoms having the same electronegativity Same electronegativity ☒ D C-F bonds are polar due to electronegativity difference of 1.4 CF₄ is a non-polar molecule due to tetrahedral shape																																																
4	C	53	① 2I⁻ → I₂ + 2e⁻ ② MnO₄⁻ + 8H⁺ + 5e⁻ → Mn²⁺ + 4H₂O ① x5 10I⁻ → 5I₂ + 10e⁻ ② x2 2MnO₄⁻ + 16H⁺ + 10e⁻ → 2Mn²⁺ + 8H₂O ①'+②' 2MnO₄⁻ + 16H⁺ + 10I⁻ → 2Mn²⁺ + 8H₂O + 5I₂ 2mol 10mol 1mol 5mol																																																
5	B	52	Sulfite is in the right hand side of the electrochemical series. The equation in the electrochemical series is reversed to become an oxidation reaction. SO₃²⁻ + H₂O → SO₄²⁻ + 2H⁺ + 2e⁻ Sulfite ions will react with a species below it on the left hand side of the electrochemical series. As sulphite ions are oxidised, it will react with an oxidising agent (the oxidising agent is reduced itself) ☒ A Iodide I⁻ ions are on the right of electrochemical series ∴ no reaction ☒ B Iodine I₂ (below SO₃²⁻ on the left) is oxidised into iodide ions: I₂ + 2e⁻ → 2I⁻ ☒ C Sulphite SO₃²⁻ ions are oxidised so sulfite ions are acting as a reducing agent ☒ D Sulfate SO₄²⁻ ions cannot react with iodide I⁻ ions as SO₄²⁻ is above I⁻ on left.																																																

6	A	56	isovaleric acid 3-methylbutanoic acid  (CH₃)₂CHCH₂COOH	is an isomer of	2-methylbutanoic acid	 CH₃CH₂CH(CH₃)COOH									
7	C	70	☐ A C ₄ H ₇ OH is unsaturated. Formula would need to be C ₄ H ₉ OH to be saturated. ☐ B C ₄ H ₇ OH is unsaturated. Formula would need to be C ₄ H ₉ OH to be saturated. ☒ C secondary alcohols oxidise to ketones with acidified dichromate solution ☐ D tertiary alcohols do not oxidise with acidified dichromate solution												
8	A	90	time = 3 minutes 20 seconds = 200 seconds $\text{rate} = \frac{1}{\text{time}} = \frac{1}{200} = 0.005 \text{ s}^{-1}$												
9	B	83	☐ A Not an α-amino acid as there are <i>two</i> carbons between -COOH and -NH ₂ groups ☒ B one carbon group between -COOH and -NH ₂ groups ∴ α-amino acid ☐ C Not an α-amino acid as there are <i>four</i> carbons between -COOH and -NH ₂ groups ☐ D Not an α-amino acid as there are <i>three</i> carbons between -COOH and -NH ₂ groups												
10	A	74	☒ A Isoprene C ₅ unit is on the end of C ₁₀ terpene ☐ B not an isoprene group as C ₅ group is in the middle of C ₁₀ terpene ☐ C not an isoprene group as group only has four carbons. Isoprene has 5 carbons. ☐ D not an isoprene group as group only has four carbons. Isoprene has 5 carbons												
11	B	61	The higher the iodine number the greater the number of C=C double bonds. • the greater the number of C=C double bonds the higher the degree of unsaturation and the further apart the edible oil molecules are • the further apart the edible oil molecules the lower the melting point .												
12	C	59	☐ A propan-2-ol produces only one product on dehydration: propene ☐ B pentan-3-ol produces only one product on dehydration: pent-2-ene ☒ C hexan-3-ol produces two products on dehydration: hex-2-ene and hex-3-ene ☐ D heptan-4-ol produces only one product on dehydration: hept-4-ene												
13	D	54	☐ A Condensation: small molecules join together with small molecule/water removed ☐ B Hydrolysis: larger molecule splits with small molecule/water added across break ☐ C Oxidation: increase in the oxygen : hydrogen ratio e.g. primary alcohol to carboxylic acid ☒ D Reduction: decrease in the oxygen : hydrogen ratio e.g. carboxylic acid to primary alcohol <table><tr><td></td><td>C₄H₈O₂</td><td>C₄H₁₀O</td></tr><tr><td>O:H ratio</td><td>2:8</td><td>1:10</td></tr><tr><td></td><td>1:4</td><td>1:5</td></tr></table>					C ₄ H ₈ O ₂	C ₄ H ₁₀ O	O:H ratio	2:8	1:10		1:4	1:5
	C ₄ H ₈ O ₂	C ₄ H ₁₀ O													
O:H ratio	2:8	1:10													
	1:4	1:5													
14	C	72	☐ A covalent bonding is inside molecules not between molecules ☐ B alkanes are non-polar ∴ no hydrogen bonding ☒ C as alkanes are non-polar, the longer the chain length the greater the strength of LDF due to the increased number of electrons in bigger molecules ☐ D alkanes are non-polar ∴ no permanent dipole to permanent dipole interactions												
15	A	67	☒ A catalysts do not increase energy of molecule to give more successful collisions ☐ B catalysts <i>do</i> lower the energy that molecules need for successful collisions ☐ C catalysts <i>do</i> provide an alternative reaction pathway to the products ☐ D catalysts <i>do</i> increase the rate of reaction												
16	D	44	Ionic formula of (NH ₄) ₃ PO ₄ = (NH ₄ ⁺) ₃ PO ₄ ³⁻ ∴ 1 mol (NH ₄) ₃ PO ₄ = 3 mol NH ₄ ⁺ positive ions ∴ 5 mol (NH ₄) ₃ PO ₄ = 15 mol NH ₄ ⁺ positive ions												
17	D	43	☐ A X-Y would give the activation energy of the forward reaction ☐ B Y-X would give a negative value and activation energies are always positive ☐ C Y-Z is the enthalpy change for the reverse reaction as it gives a positive value ☒ D Z-Y is the enthalpy change for the forward reaction as it gives a negative value												

18	B	75	$\begin{array}{ccc} & \xrightarrow{X} & \\ \text{C(g)} + \text{O}_2\text{(g)} & & \text{C(g)} + \text{O}_2\text{(g)} \\ & \searrow \Delta H_1 = -393.5 \text{ kJ mol}^{-1} & \swarrow \Delta H_2 = -283.0 \text{ kJ} \\ & \text{CO}_2\text{(g)} & \end{array}$ $X = \Delta H_1 - \Delta H_2$ $X = -393.5 - (-283.0)$ $X = -110.5 \text{ kJ mol}^{-1}$
19	A	47	☒ A 3xN-H bonds formed in reaction $\therefore \frac{1}{3}$ of total is enthalpy for 1xN-H bond ☒ B Bonds in N_2 and H_2 must be broken before N-H bonds in NH_3 are formed ☒ C Bonds in N_2 and H_2 must be broken before N-H bonds in NH_3 are formed ☒ D Bonds in H_2 must be broken before N-H bonds in NH_3 are formed
20	D	83	As the reaction is already at equilibrium and the conditions have not changed, the rate of the forward reaction and reverse reactions will be the same.
21	A	79	$\textcircled{1}$ $\textcircled{2}$ $\textcircled{3}$ $\textcircled{1} \times -1$ $\textcircled{2}$ $\textcircled{3}$ Add $\textcircled{1} + \textcircled{2} + \textcircled{3}$ $\text{HCOOH} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{HCOOH} + \frac{1}{2}\text{O}_2$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\text{C} + \text{H}_2 + \text{O}_2 \rightarrow \text{HCOOH}$ $\Delta H = b$ $\Delta H = c$ $\Delta H = d$ $\Delta H = -b$ $\Delta H = c$ $\Delta H = d$ $\Delta H = c + d - b = a$
22	B	42	☒ A X is the residue which gets trapped in the filter paper during filtration ☒ B X is the residue and Y is the filtrate ☒ C X is the residue which gets trapped in the filter paper during filtration ☒ D Y is the filtrate which goes through the filter paper during filtration
23	C	41	☒ A 50cm ³ could not be measured accurately in a beaker ☒ B 50cm ³ pipette can only measure exactly 50cm ³ ☒ C 45cm ³ of solution could be measured accurately in a 50cm ³ burette ☒ D 45cm ³ could not be measured as accurately in a measuring cylinder as in a burette
24	B	86	☒ A Repeating the titration is necessary but not for observing the end-point ☒ B A white tile helps observation of exact point where colour change takes place ☒ C Rinsing the flask between is necessary but not for observing the end-point ☒ D Disregarding the rough titre is necessary but not for observing the end-point
25	B	48	☒ A Bubbling CO_2 through calcium hydroxide (lime water) would remove the CO_2 but CO_2 would not be able to be collected. ☒ B Collection over water would isolate CO_2 from the reactants & other products ☒ C CO_2 is already a gas so evaporation is not suitable ☒ D Distillation separates liquids with different boiling points

2024 Higher Chemistry Marking Scheme

Long Qu	Answer	Reasoning																		
1a(i)	Answer to include:	Energy required to remove 1 mole of electrons from 1 mole of atoms in the gaseous state.																		
1a(ii)	$P(g) \rightarrow P^+(g) + e^-$	1 st Ionisation energy is the removal of one mole of electrons and. Turns gaseous atoms into gaseous one positive ions. Must be in the gaseous state.																		
1b	Answer to include:	<table><tr><td>1 mark</td><td>Stronger</td><td> intermolecular van der Waals' </td><td>forces between phosphorus compared with nitrogen</td></tr><tr><td>1 mark</td><td colspan="3">London dispersion forces (LDFs) are the intermolecular forces present.</td></tr><tr><td>1 mark</td><td colspan="3">There are more electrons in phosphorus (P₄) compared to nitrogen (N₂)</td></tr></table>	1 mark	Stronger	intermolecular van der Waals'	forces between phosphorus compared with nitrogen	1 mark	London dispersion forces (LDFs) are the intermolecular forces present.			1 mark	There are more electrons in phosphorus (P ₄) compared to nitrogen (N ₂)								
1 mark	Stronger	intermolecular van der Waals'	forces between phosphorus compared with nitrogen																	
1 mark	London dispersion forces (LDFs) are the intermolecular forces present.																			
1 mark	There are more electrons in phosphorus (P ₄) compared to nitrogen (N ₂)																			
1c	230	<table><tr><th colspan="3">Bond Breaking Steps</th><th colspan="3">Bond Forming Steps</th></tr><tr><th>No. of bonds</th><th>Energy</th><th>Total</th><th>No. of bonds</th><th>Energy</th><th>Total</th></tr><tr><td>6x P-P</td><td>6x 201</td><td>1206 kJ</td><td>2x P≡P</td><td>2x 488</td><td>976 kJ</td></tr></table> Enthalpy change = ΣBond breaking steps - Σbond forming steps Enthalpy change = 1206kJ - 976kJ Enthalpy change = 230kJ mol⁻¹	Bond Breaking Steps			Bond Forming Steps			No. of bonds	Energy	Total	No. of bonds	Energy	Total	6x P-P	6x 201	1206 kJ	2x P≡P	2x 488	976 kJ
Bond Breaking Steps			Bond Forming Steps																	
No. of bonds	Energy	Total	No. of bonds	Energy	Total															
6x P-P	6x 201	1206 kJ	2x P≡P	2x 488	976 kJ															
1d(i)	77.1%	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{750}{310.3} = 2.42 \text{ mol}$ $2\text{Ca}_3(\text{PO}_4)_2 + 6\text{SiO}_2 + 10\text{C} \longrightarrow 6\text{CaSiO}_3 + 10\text{CO} + \text{P}_4$ 2mol 2.42mol 1mol 1.21mol mass = no. of mol x gfm = 1.21 mol x 124.0 = 149.9g (theoretical) % Yield = $\frac{\text{actual}}{\text{theoretical}} \times 100 = \frac{115.5}{149.9} \times 100 = 77.1\%$																		
1d(ii)	reducing agent	A reducing agent is a species which reduces another species in a reaction while being oxidised itself																		
1d(iii)	147 litres	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{280}{28.0} = 10.0 \text{ mol}$ Volume = 1450 litres Molar Volume = $\frac{\text{Volume}}{\text{no. of mol}} = \frac{1470}{10.0} = 147 \text{ litres mol}^{-1}$																		
2a(i)	activation energy	The activation energy is the minimum energy required in a reaction to form an activated complex and then form products.																		
2a(ii)	-52.4	gfm NaN₃ = 65.0g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{80}{65.0} = 1.23 \text{ mol}$ 1 mol NaN₃ releases -42.6kJ 1.23 mol NaN₃ releases -42.6kJ x ^{1.23}/₁ = -52.4 kJ																		
2a(iii)	one negative or 1-	As Na carries a one positive charge as a Na ⁺ ion, the negative ion must carry a one negative charge making the azide group ion N ₃ ⁻																		
2b	balanced equation showing:	$10\text{Na} + 2\text{KNO}_3 \longrightarrow \text{K}_2\text{O} + 5\text{Na}_2\text{O} + \text{N}_2$																		
2c	Answer to include:	<table><tr><td>1 mark</td><td>Silicon dioxide is a covalent network</td></tr><tr><td>1 mark</td><td>(Strong) covalent bonds in network are broken</td></tr></table>	1 mark	Silicon dioxide is a covalent network	1 mark	(Strong) covalent bonds in network are broken														
1 mark	Silicon dioxide is a covalent network																			
1 mark	(Strong) covalent bonds in network are broken																			

2d	47.3	$\text{gfm NaN}_3 = 65.0\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{80}{65.0} = 1.23 \text{ mol}$ sodium azide $\longrightarrow$ nitrogen 1mol 1.6mol 1.23mol 1.6mol $\times 1.23/1$ = 1.97mol Volume = no. of mol $\times$ Molar Volume = 1.97 $\times$ 24 = 47.3 litres					
2e	Answer to include:	1 mark	powders have large/greater surface area				
		1 mark	more successful collisions				
3	Open Question Answer to Include:	3 mark answer		2 mark answer		1 mark answer	
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.	
		Candidate Marks:					
		3 marks	2 marks	1 mark	0 marks		
		11%	27%	34%	29%		
4a(i)	Water	The joining together of amino acids to form proteins is a condensation reaction. In a condensation reaction, a small molecule is removed and in the formation of proteins, water is removed as the amino acids join together.					
4a(ii)	One answer from:	Answer	Oxidation		Condensation		Dehydrogenation
		Reasoning	Increase in oxygen : hydrogen ratio by removal of hydrogen		Joining together with small molecule removed at join		Removal of hydrogen reaction
4a(iii)	An amino acid which can only be obtained through your diet.	Essential amino acids are the amino acids the body needs to make a range of proteins but the essential amino acids have to be taken into the body via your diet and cannot be made in the body.					
4a(iv)A	Denaturing	Denaturing occurs when the shape of a protein/enzyme is irreversibly changed and will not return to its original shape.					
4a(iv)B	one answer from:	$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \\ \text{H}-\text{N}-\text{C}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$ $\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \\ \text{H}-\text{N}-\text{C}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{O} \\ \\ \text{H} \end{array}$ $\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \\ \text{H}-\text{N}-\text{C}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{S} \\ \\ \text{H} \end{array}$					
4a(iv)C	One dotted line as shown:	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{O} & \text{H} & \text{H} & \text{O} & \text{H} & \text{H} & \text{O} \\ & & & & & & & & \\ -\text{N}-\text{C}-\text{C}-\text{N}-\text{C}-\text{C}-\text{N}-\text{C}-\text{C}- \\ & & & & & & & & \\ \text{H} & & & \text{H}-\text{C}-\text{H} & & & \text{H}-\text{C}-\text{H} & & \\ & & & & & & & & \\ & & & \text{O} & & & \text{S} & & \\ & & & & & & & & \\ & & & \text{H} & & & \text{H} & & \end{array}$ or $\begin{array}{ccccccc} \text{H} & \text{H} & \text{O} & \text{H} & \text{H} & \text{O} & \text{H} & \text{H} & \text{O} \\ & & & & & & & & \\ -\text{N}-\text{C}-\text{C}-\text{N}-\text{C}-\text{C}-\text{N}-\text{C}-\text{C}- \\ & & & & & & & & \\ \text{H} & & & \text{H}-\text{C}-\text{H} & & & \text{H}-\text{C}-\text{H} & & \\ & & & & & & & & \\ & & & \text{O} & & & \text{S} & & \\ & & & & & & & & \\ & & & \text{H} & & & \text{H} & & \end{array}$ $\begin{array}{c} \text{H} & \text{O} \\ & \\ \text{H}-\text{O} & -\text{H} \end{array}$					
4b(i)	0.467	$R_f = \frac{\text{distance travelled by amino acid}}{\text{distance travelled by solvent front}} = \frac{5.6}{12.0} = 0.467$					
4b(ii)	One answer from:	The spots for glycine and lysine overlap/are close together			The R_f values are similar/close together for glycine and lysine.		
4b(iii)	Threonine	Threonine is the most polar as it is the only amino acid shown with a polar -OH hydroxyl group on the side group. Each of the amino acids have polar -NH ₂ and -COOH groups so the polar effect of these groups cancel each other out.					
5a	One answer from:	Fats are a concentrated source of energy			Fats are essential for the transport/storage of fat-soluble vitamins in the body		

6a(i)	methyl octanoate	$\text{CH}_3(\text{CH}_2)_6\text{COOCH}_3$ has the structure: 8 carbon carboxylic acid side (2nd part of ester name) 1 carbon alcohol side (1st part of ester name) = -octanoate = methyl-																																					
6a(ii)A	answer to include:	<table><tr><td>1 mark</td><td>Diagram shows a labelled safe method of heating (no flames) and labelled reaction mixture.</td></tr><tr><td>1 mark</td><td>Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube</td></tr></table>	1 mark	Diagram shows a labelled safe method of heating (no flames) and labelled reaction mixture.	1 mark	Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube	cotton wool wet paper towel elastic band test tube beaker hot water alcohol + carboxylic acid + catalyst																																
1 mark	Diagram shows a labelled safe method of heating (no flames) and labelled reaction mixture.																																						
1 mark	Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube																																						
6a(ii)B	diagram showing:	The peak drawn with dotted line shows a higher activation energy. potential energy reaction pathway The line drawn by the candidate must start and finish at the same potential energies and positions on the reaction pathway.																																					
6b(i)	One set from:	<table><tr><td>1 mark</td><td colspan="3">Correctly identifies molecule has two parts with different polarities</td></tr><tr><td>1 mark</td><td>Correctly identifies</td><td> polar part hydroxyl groups </td><td> dissolves in water dissolves in polar liquids are hydrophilic </td></tr><tr><td></td><td></td><td>and</td><td></td></tr><tr><td></td><td>Correctly identifies</td><td> non-polar part hydrocarbon chain </td><td> dissolves in biodiesel dissolves in non-polar liquids are hydrophobic </td></tr><tr><td></td><td></td><td colspan="2">or</td></tr><tr><td>1 mark</td><td colspan="3">Correctly identifies molecule has two parts as hydrophilic and hydrophobic.</td></tr><tr><td>1 mark</td><td>Correctly identifies</td><td>hydrophilic part</td><td> dissolves in water dissolves in polar liquids </td></tr><tr><td></td><td></td><td>and</td><td></td></tr><tr><td></td><td>Correctly identifies</td><td>hydrophobic part</td><td> dissolves in biodiesel dissolves in non-polar liquids </td></tr></table>		1 mark	Correctly identifies molecule has two parts with different polarities			1 mark	Correctly identifies	polar part hydroxyl groups	dissolves in water dissolves in polar liquids are hydrophilic			and			Correctly identifies	non-polar part hydrocarbon chain	dissolves in biodiesel dissolves in non-polar liquids are hydrophobic			or		1 mark	Correctly identifies molecule has two parts as hydrophilic and hydrophobic.			1 mark	Correctly identifies	hydrophilic part	dissolves in water dissolves in polar liquids			and			Correctly identifies	hydrophobic part	dissolves in biodiesel dissolves in non-polar liquids
1 mark	Correctly identifies molecule has two parts with different polarities																																						
1 mark	Correctly identifies	polar part hydroxyl groups	dissolves in water dissolves in polar liquids are hydrophilic																																				
		and																																					
	Correctly identifies	non-polar part hydrocarbon chain	dissolves in biodiesel dissolves in non-polar liquids are hydrophobic																																				
		or																																					
1 mark	Correctly identifies molecule has two parts as hydrophilic and hydrophobic.																																						
1 mark	Correctly identifies	hydrophilic part	dissolves in water dissolves in polar liquids																																				
		and																																					
	Correctly identifies	hydrophobic part	dissolves in biodiesel dissolves in non-polar liquids																																				
6b(ii)	Reacting edible oils with glycerol	glycerol edible oil emulsifier + H₂O																																					

6c(i)	One answer from:	<table><tr><td>size of wick</td><td>use same draught shield</td><td>material/type of container</td></tr><tr><td>stirring water in both</td><td>distance between burner/wick/flame & container</td><td>indication of same insulation in both experiments</td></tr></table>	size of wick	use same draught shield	material/type of container	stirring water in both	distance between burner/wick/flame & container	indication of same insulation in both experiments		
size of wick	use same draught shield	material/type of container								
stirring water in both	distance between burner/wick/flame & container	indication of same insulation in both experiments								
6c(ii)	One answer from:	<table><tr><td>use a draught shield</td><td>move flame closer to container</td><td>fit a lid on the container</td><td>insulate container</td></tr></table>	use a draught shield	move flame closer to container	fit a lid on the container	insulate container				
use a draught shield	move flame closer to container	fit a lid on the container	insulate container							
6c(iii)	41.8 or -41.8	Change in Temperature = $38^{\circ}\text{C} - 17^{\circ}\text{C} = 21^{\circ}\text{C}$ Heat Energy = Specific Heat Capacity $\times$ Mass $\times$ Change In Temperature $E_h = c \times m \times \Delta T$ $E_h = 4.18 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \times 0.2\text{kg} \times 21^{\circ}\text{C}$ $E_h = 17.556 \text{ kJ}$ 0.420g biodiesel $\longleftrightarrow$ 17.556 kJ 1g biodiesel $\longleftrightarrow$ $17.556 \text{ kJ} \times \frac{1}{0.420}$ $= 41.8 \text{ kJ g}^{-1}$								
6d	£12.75	20% of 75 litres of E20 = $\frac{20}{100} \times 75 \text{ litres} = 15 \text{ litres of biodiesel.}$ 1 litre biodiesel $\longleftrightarrow$ £0.85 15 litres biodiesel $\longleftrightarrow$ $£0.85 \times \frac{15}{1} = £12.75$								
7a	19.4	Salinity = Chloride Ion concentration $\times$ 0.0018066 Chloride Ion concentration = $\frac{\text{Salinity}}{0.0018066}$ Chloride Ion concentration = $\frac{35}{0.0018066}$ Chloride Ion concentration = 19373 mg l^{-1} $= 19.4 \text{ g l}^{-1}$								
7b(i)A	One from:	Solution of $\left\{ \begin{array}{l} \text{accurately} \\ \text{precisely} \\ \text{exactly} \end{array} \right\}$ known concentration								
7b(i)B	Answer to include:	<table><tr><td>1 mark</td><td colspan="3">Rinse with seawater/sample/solution</td></tr><tr><td>1 mark</td><td>Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)</td><td>or</td><td>Fill to mark, read from the bottom of the meniscus</td></tr></table>	1 mark	Rinse with seawater/sample/solution			1 mark	Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)	or	Fill to mark, read from the bottom of the meniscus
1 mark	Rinse with seawater/sample/solution									
1 mark	Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)	or	Fill to mark, read from the bottom of the meniscus							
7b(ii)	To identify the end point	An indicator is used to show the end point of a reaction. The colour change of the indicator can literally be caused by one drop of reactant so care is needed when approaching the end point. (Some reactions are self-indicating and need no indicator e.g. acidified potassium permanganate)								
7b(iii)A	One answer from:	<table><tr><td>only concordant values are used in average</td><td>sample 1 is not concordant.</td></tr></table>	only concordant values are used in average	sample 1 is not concordant.						
only concordant values are used in average	sample 1 is not concordant.									
7b(iii)B	0.5775	no. of mol = volume $\times$ concentration = $0.01155\text{litres} \times 0.5\text{mol l}^{-1} = 0.005775\text{mol}$ Ag^+ ion react with Cl^- ions <table><tr><td>1mol</td><td></td><td>1mol</td></tr><tr><td>0.005775mol</td><td></td><td>0.005775mol</td></tr></table> 10.0cm³ of seawater contains 0.005775mol Cl⁻ ions 1000cm³ of seawater contains $0.005775\text{mol Cl}^- \text{ ions} \times \frac{1000}{10.0}$ $= 0.5775 \text{ mol l}^{-1}$	1mol		1mol	0.005775mol		0.005775mol		
1mol		1mol								
0.005775mol		0.005775mol								
7b(iii)C	17.47	10cm³ of different seawater contains 0.00492mol Cl⁻ ions 1000cm³ of different seawater contains $0.00492\text{mol Cl}^- \text{ ions} \times \frac{1000}{10}$ $= 0.492\text{mol Cl}^- \text{ ions}$ mass. = no. of mol $\times$ gfm = $0.492 \times 35.5 = 17.466\text{g}$								

7c	Answer to include:	1 mark	Chloromethane has permanent dipole to permanent dipole interactions whereas water has hydrogen bonding		
		1 mark	Hydrogen bonding is stronger than permanent dipole to permanent dipole interactions		
8	Open Question:	3 mark answer		2 mark answer	1 mark answer
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
		Candidate Marks:			
		3 marks	2 marks	1 mark	0 marks
		10%	21%	26%	42%
9a(i)	2843.55	mass of $\text{TiO}_2 = 1600\text{kg} = 1600000\text{g}$ gfm $\text{TiO}_2 = 79.9$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1600000}{79.9} = 20025 \text{ mol}$ $\begin{array}{ccccccc} \text{TiO}_2 & + & 2\text{Cl}_2 & + & \text{C} & \longrightarrow & \text{TiCl}_4 + \text{CO}_2 \\ 1\text{mol} & & 2\text{mol} & & & & \\ 20025\text{mol} & & 40050\text{mol} & & & & \end{array}$ gfm $\text{Cl}_2 = 71.0$ mass = no. of mol $\times$ gfm = $40050 \times 71.0 = 2843554\text{g}$ = 2843.554kg			
9a(ii)	Covalent molecular	If Titanium Chloride were to follow the convention that ionic compounds are formed when a metal and non-metal combine then titanium would be a solid at room temperature. Because titanium chloride is a liquid at room temperature then we can conclude that titanium chloride is covalent molecular.			
9b(i)	20.1%	atom economy = $\frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{(1 \times 47.9)}{(1 \times 189.9) + (2 \times 24.3)} \times 100 = 20.1\%$			
9b(ii)	$\text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^-$	This oxidation reaction is on p13 of data booklet but reversed as a reduction reaction. Oxidation reactions have electrons after the arrow Magnesium forms a $2+$ ion as magnesium atoms have electron arrangement of 2,8,2 and two electrons are lost to become the Mg^{2+} ion with the stable electron arrangement of 2,8.			
9b(iii)	10854	mass of Mg available = $750\text{kg} = 750000\text{g}$ gfm Mg = 24.3 no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{750000}{24.3} = 30864 \text{ mol}$ mass of $\text{TiCl}_4 = 1900\text{kg} = 1900000\text{g}$ gfm Mg = 189.9 no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1900000}{189.9} = 10005 \text{ mol}$ $\begin{array}{ccccccc} \text{TiCl}_4 & + & 2\text{Mg} & \longrightarrow & \text{Ti} & + & \text{MgCl}_2 \\ 1\text{mol} & & 2\text{mol} & & & & \\ 10005\text{mol} & & 20010\text{mol (reacted)} & & & & \end{array}$ no. of mol Mg unreacted = $30864\text{mol} - 20010\text{mol} = 10854\text{mol Mg}$			
9b(iv)	Argon is unreactive	Any oxygen present in this reaction is going to react vigorously with magnesium so replacing air with argon removes the possibility of magnesium reacting with oxygen			
9c	One answer from:		can be carried out at lower temperature	requires less energy	has a higher atom economy

10a	6.0 - 4.5	From p17 of data booklet: <table><thead><tr><th>Chemical Group</th><th>Chemical Shift Values</th></tr></thead><tbody><tr><td>$\text{-}\overset{ }{\text{C}}=\text{CH}_2$, $\text{-}\overset{ }{\text{C}}=\overset{ }{\text{CH}}$</td><td> 6.0-4.5 </td></tr></tbody></table>	Chemical Group	Chemical Shift Values	$\text{-}\overset{ }{\text{C}}=\text{CH}_2$, $\text{-}\overset{ }{\text{C}}=\overset{ }{\text{CH}}$	6.0-4.5											
Chemical Group	Chemical Shift Values																
$\text{-}\overset{ }{\text{C}}=\text{CH}_2$, $\text{-}\overset{ }{\text{C}}=\overset{ }{\text{CH}}$	6.0-4.5																
10b	propanone (or acetone)	Possible structures of $\text{C}_3\text{H}_6\text{O}$ are: <table><thead><tr><th>Propanal</th><th>Propanone</th></tr></thead><tbody><tr><td> </td><td> </td></tr><tr><td> There should be 2 vertical lines for propanal corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-R}$</td><td>$\text{-CH}_2\text{-CHO}$</td></tr><tr><td>1.5-0.9</td><td>2.7-2.0</td></tr></tbody></table></td><td> There should be 1 vertical line for propanone corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-C=O}$</td><td>O=C-CH_3</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></tbody></table> There is only one vertical line as the two side are identical. </td></tr></tbody></table>	Propanal	Propanone			There should be 2 vertical lines for propanal corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-R}$</td><td>$\text{-CH}_2\text{-CHO}$</td></tr><tr><td>1.5-0.9</td><td>2.7-2.0</td></tr></tbody></table>	$\text{CH}_3\text{-R}$	$\text{-CH}_2\text{-CHO}$	1.5-0.9	2.7-2.0	There should be 1 vertical line for propanone corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-C=O}$</td><td>O=C-CH_3</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></tbody></table> There is only one vertical line as the two side are identical.	$\text{CH}_3\text{-C=O}$	O=C-CH_3	2.7-2.0	2.7-2.0	
Propanal	Propanone																
There should be 2 vertical lines for propanal corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-R}$</td><td>$\text{-CH}_2\text{-CHO}$</td></tr><tr><td>1.5-0.9</td><td>2.7-2.0</td></tr></tbody></table>	$\text{CH}_3\text{-R}$	$\text{-CH}_2\text{-CHO}$	1.5-0.9	2.7-2.0	There should be 1 vertical line for propanone corresponding to <table><tbody><tr><td>$\text{CH}_3\text{-C=O}$</td><td>O=C-CH_3</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></tbody></table> There is only one vertical line as the two side are identical.	$\text{CH}_3\text{-C=O}$	O=C-CH_3	2.7-2.0	2.7-2.0								
$\text{CH}_3\text{-R}$	$\text{-CH}_2\text{-CHO}$																
1.5-0.9	2.7-2.0																
$\text{CH}_3\text{-C=O}$	O=C-CH_3																
2.7-2.0	2.7-2.0																
10c		<table><thead><tr><th>Structure</th><th colspan="2"> </th></tr></thead><tbody><tr><td>Chemical Group</td><td>$\text{CH}_3\text{-R}$</td><td>$\text{R-CH}_2\text{-X}$</td></tr><tr><td>Chemical Shift</td><td>1.5-0.9</td><td>4.2-2.2</td></tr><tr><td>No of hydrogens</td><td>3</td><td>2</td></tr><tr><td>Relative Height</td><td>3</td><td>2</td></tr></tbody></table>	Structure			Chemical Group	$\text{CH}_3\text{-R}$	$\text{R-CH}_2\text{-X}$	Chemical Shift	1.5-0.9	4.2-2.2	No of hydrogens	3	2	Relative Height	3	2
Structure																	
Chemical Group	$\text{CH}_3\text{-R}$	$\text{R-CH}_2\text{-X}$															
Chemical Shift	1.5-0.9	4.2-2.2															
No of hydrogens	3	2															
Relative Height	3	2															